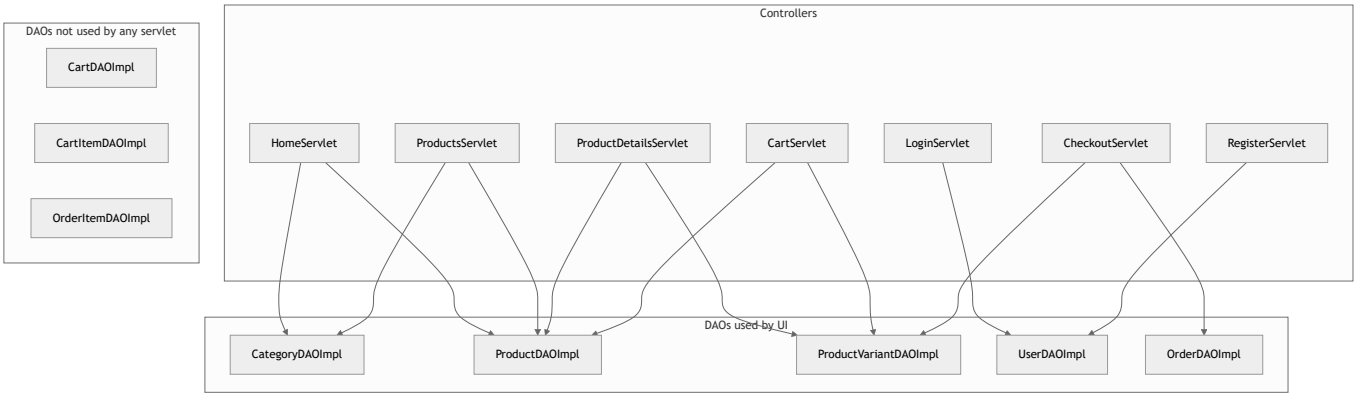
Controllers and DAO usage

Servlets hold DAO fields as interfaces and construct `*DAOImpl` in `init()` or as instance fields. There is no factory, Spring, or service class.

Wiring diagram



URL map

Mapping	Class	Methods
`/home`	`HomeServlet`	GET
`/products`	`ProductsServlet`	GET
`/product`	`ProductDetailsServlet`	GET
`/login`	`LoginServlet`	GET, POST
`/register`	`RegisterServlet`	GET, POST
`/cart`	`CartServlet`	GET, POST (`action` = add / update / remove)
`/checkout` , `/checkout/confirmation`	`CheckoutServlet`	GET, POST

Mappings come from `@WebServlet` , not `web.xml` .

`HomeServlet`

DAOs: `CategoryDAO` , `ProductDAO`

Method	DAO call
`GET`	`categoryDAO.getAllCategories()`
`GET`	`productDAO.getLatestProducts(8)`

Attributes: `categories`, `latestProducts` → `home.jsp`.

`ProductsServlet`

DAOs: `ProductDAO`, `CategoryDAO`

Method	DAO call
`GET`	`categoryDAO.getAllCategories()`
`GET`	`productDAO.getFilteredProducts(keyword, categoryId, min, max, null, null, null, sortBy, "ASC", 1, 100)`
`GET`	`productDAO.countFilteredProducts(keyword, categoryId, min, max, null, null, null)`

Unused DAO surface from this servlet: brand/color/size filters, other sort helpers, `getProductsByPage`.

`ProductDetailsServlet`

DAOs: `ProductDAO`, `ProductVariantDAO`

Method	DAO call
`GET`	`productDAO.getProductById(id)`
`GET`	`productVariantDAO.getVariantsByProductId(id)`

Redirects to `/products` if `id` is missing. 404 if product is null. Does not call `getRelatedProducts`.

`LoginServlet`

DAOs: `UserDAO`

Method	DAO call
`GET`	none
`POST`	`userDAO.loginUser(email, password)`

Session after success: `user` (`User`), `userId` (`int`). Old session invalidated first.

`CheckoutServlet` also accepts `loggedInUser`; login never sets that name.

`RegisterServlet`

DAOs: `UserDAO`

Method	DAO call
`GET`	none
`POST`	`userDAO.emailExists(email)`
`POST`	`userDAO.phoneExists(phone)` if phone non-blank
`POST`	`userDAO.registerUser(user)`

Does not auto-login. Redirects to `/login?registered=true`.

`CartServlet`

DAOs: `ProductDAO`, `ProductVariantDAO`
Not used: `CartDAO`, `CartItemDAO`

Method	DAO call	Storage
`GET`	none	session `cart`
`POST add`	`productDAO.getProductById`	session
`POST add`	`variantDAO.getVariantById`	session
`POST add`	`variantDAO.isStockAvailable`	session
`POST update`	`variantDAO.isStockAvailable`	session
`POST remove`	none	session

Cart key: `productId + "-" + variantId`. Guest carts work (no login required).

`CheckoutServlet`

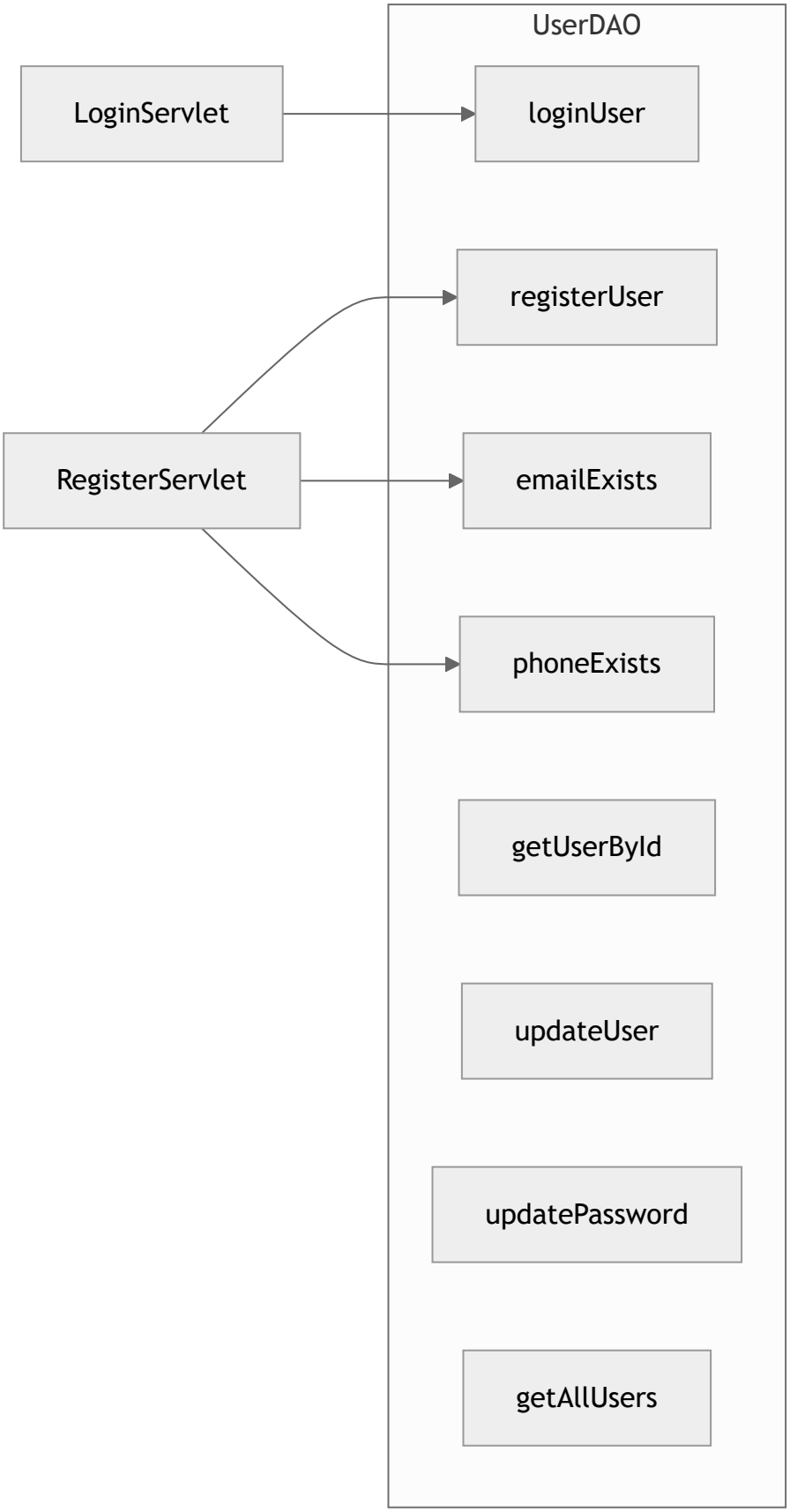
DAOs: `OrderDAO`, `ProductVariantDAO`
Not used: `OrderItemDAO`, `UserDAO`, `CartDAO`

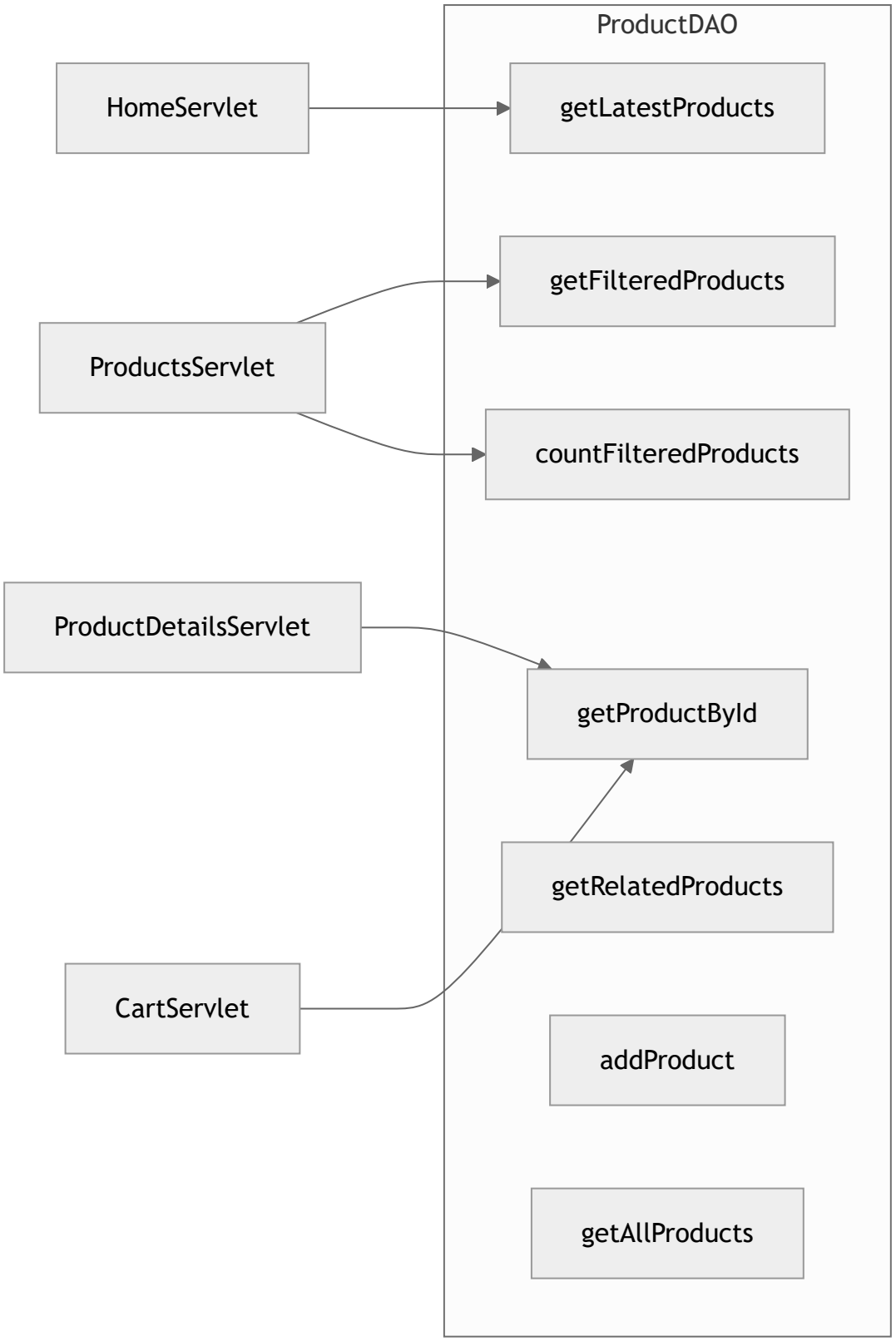
Method	Path	DAO call
`GET`	`/checkout`	none (session user + cart)
`GET`	`/checkout/confirmation`	none (`lastOrder`)
`POST`	`/checkout`	`variantDAO.isStockAvailable` per line
`POST`	`/checkout`	`orderDAO.placeOrder(order)`

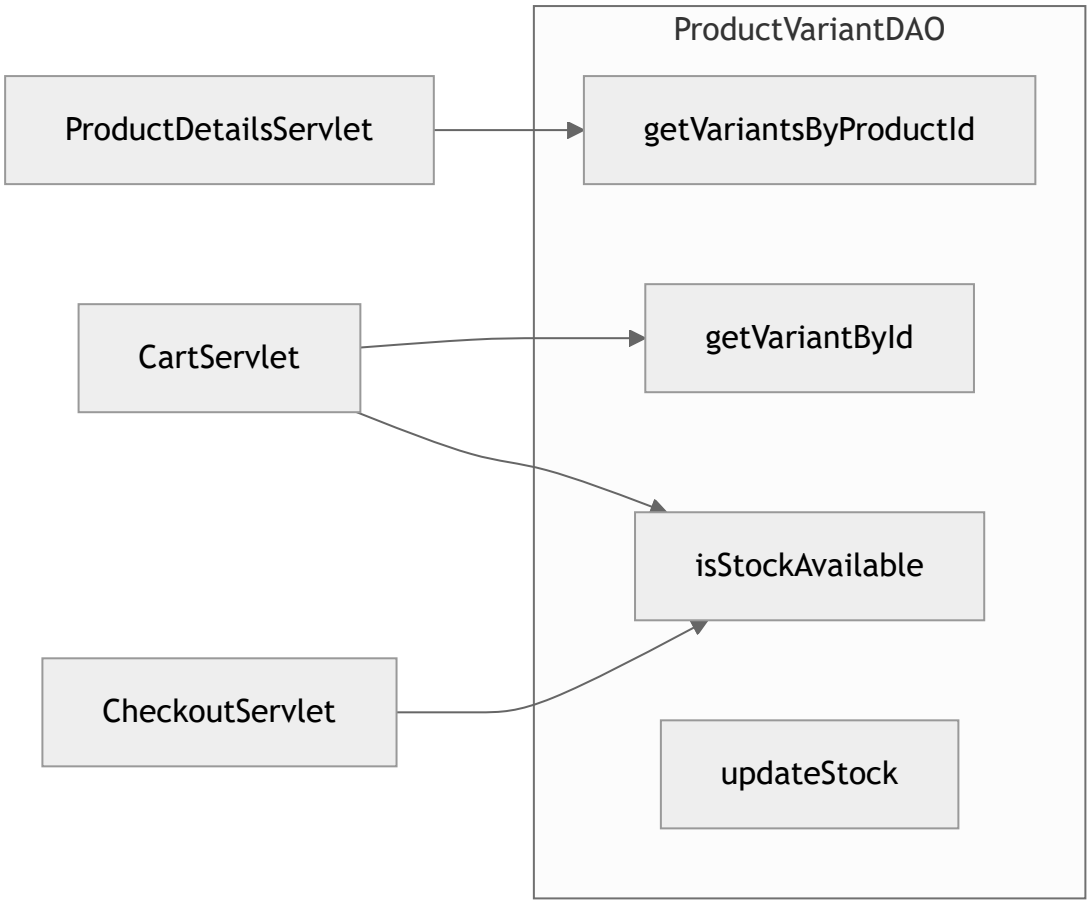
Gate: must have session `user` or `loggedInUser`. Payment must be `COD` (stored as `"Cash on Delivery"`). Status written as `"PENDING"`.

After success: `lastOrder`, `lastOrderItems` (copy of session cart), then `cart` removed. No insert into `order_items`, no `updateStock`.

DAO methods vs servlet callers




ProductDAO
HomeServlet
getLatestProducts
getFilteredProducts
ProductsServlet
countFilteredProducts
ProductDetailsServlet
getProductById
getRelatedProducts
CartServlet
addProduct
getAllProducts



Methods with **no servlet caller** (available for admin/tests): `UserDAO` get/update/list; `CategoryDAO` add/update/delete/getById; most `ProductDAO` CRUD/sort helpers; `ProductVariantDAO` CRUD/ `updateStock` ; entire `CartDAO` , `CartItemDAO` , `OrderItemDAO` ; `OrderDAO` get/list/status/cancel.

Session contract (not a table)

Attribute	Set by	Read by
`user`	`LoginServlet`	`CheckoutServlet` , JSPs
`userId`	`LoginServlet`	unused by other servlets
`cart`	`CartServlet`	`CartServlet` , `CheckoutServlet` , `navbar.jsp`
`lastOrder`	`CheckoutServlet` POST	confirmation GET + JSP
`lastOrderItems`	`CheckoutServlet` POST	`order-confirmation.jsp`

Gaps the team should treat as current design, not accidents

1. DB cart tables are implemented; UI cart is session-only (lost on timeout/new device; not tied to `users`).
2. Checkout persists order header only; line items live in session until the browser session ends.
3. Stock is validated, never reduced.

4. No logout, no order-history servlet, no admin product/category UI.